KARNATAK UNIVERSITY, DHARWAD
DEPARTMENT OF STUDIES IN STATISTICS
2024-25

PROJECT REPORT ON

" EXCESS USE OF MOBILE AND MENTAL HEALTH: A STATISTICAL ANALYSIS"

SUBMITTED BY:

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MASTER OF SCIENCE IN "STATISTICS"

PROJECT GUIDE

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CHAPTER 1
INTRODUCTION

SOCIALMEDIA is the latest type of media and have additionally acquired dumbfounding overall development. It has become a piece of everybody's life. Web based media is a wide term and truly includes a few unique sorts of media, like recordings, web journals, and so forth it is where you can send data about others.

Online media has been a significant piece of one's life from shopping to electronic sends, instruction and business devices. Online media assumes an essential part in changing individuals' way of life. Online media incorporates person to person communication locales and websites where individuals can without much of a stretch associate with each other. Since the development of these interpersonal interaction destinations like Twitter and Facebook as key instruments for news, writers and their associations have played out a difficult exercise. These destinations have gotten per everyday daily practice for individuals.

Web-based media has been essentially characterized to allude to "the numerous generally cheap and broadly open electronic devices that encourage anybody to distribute and get to data, team up on a typical exertion, or assemble relationship" Social media is characterized as sites and applications that empower clients to make and share content or to take part in informal communication. It is the quite possibly the most current and most loved type of Social media including numerous highlights and Social-qualities in it.

It has numerous benefits on the same channel like conveying, messaging, pictures sharing sound and video sharing, quick distributing, connecting with all over the world, direct associating. It is likewise a least expensive quick admittance to the world so it is exceptionally fundamental for all age gathering of people groups. Web use is expanding step by step nowadays with high rates on the whole preposterous. Dominant part of youth is moving rapidly from electronic media like as watchers at home and radio audience members to the online media among all periods of gathering individuals. Presently a day's youth rate is especially moving into web-based media so its effects

1.1 Objectives of the Study:

1. To study the factors influencing the usage of social media among youth population.
2. To analyse the purpose of usage of social media among youth population.
3. To understand the impact of social media on youth population
4. To examine the rating given to social media platform to the youth population.
5. To examine the opinion of the youth on issues and disadvantages of using social media

CHAPTER 2

METHODOLOGY

2.1 DATA COLLECTION

A sample survey was conducted to collect data from Karnatak University Campus faculties such as Science, Commerce, Arts and Social science. It was decided to collect primary data through questionnaire.

The collected data is analysed statistically to interpret findings and results. The questionnaire contains 18 questions which include socio-demographic variable, knowledge and awareness related questions. The research is qualitative in nature.

Primary data refers to data that is collected firsthand by the researcher for a specific purpose or study. It is original, not previously collected or published, and is usually gathered through direct contact with the subjects.

- Definition:

Primary data collection is the process of gathering new, original data directly from sources such as people, observations, or experiments, specifically for the purpose of the current research.

- Characteristics:

First-hand information

Specific to the study

Accurate and up-to-date

More time-consuming and costly than secondary data

Methods of Primary Data Collection:

1. Surveys and Questionnaires

Structured forms with predefined questions

Can be paper-based or online

Example: Google Forms, printed surveys

2. Interviews

One-on-one verbal communication

Can be structured, semi-structured, or unstructured

2.2 SAMPLING TECHNIQUES

Samples drawn from all the faculties through simple random sampling. Simple Random Sampling (SRS) is a basic method of selecting a sample from a larger population where each member has an equal chance of being selected. It ensures that the sample is unbiased and representative of the population.

Definition:

Simple random sampling is a sampling technique in which every item or individual in the population has an equal probability of being chosen.

Key Features:

- Each selection is made independently.
- No favoritism or grouping is involved.
- Typically done using lottery method or random number generators.

Methods of SRS:

1. Lottery Method – Numbers are assigned to each individual, and random draws are made.

2. Random Number Table/Software – Use of tools like Excel, R, Python, or statistical calculators.

2.3 CHI-SQUARE TEST FOR INDEPENDENCE OF ATTRIBUTES

The chi-square(χ^2) test is a widely used statistical method for assessing the relationships between categorical variables. It is especially useful for determining whether observed frequencies differ significantly from expected frequencies under a specific hypothesis. There are two types of chi-square test :The chi-square goodness of fit test and the chi-square test of independence.

Chi-square test for independence formula :

$$E_{ij}=\frac{\text{row total}*\text{column total}}{\text{Grand total}}$$

\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}

Degrees of Freedom: df = (number of rows - 1) *(number of columns -1)

Terminology of the Chi-Square Test:

a. Non-Parametric: Does not assume a normal distribution of the data.
b. Categorical Data: Suitable for categorical (nominal) data.
c. Contingency Tables: Often presented in contingency tables, which display the frequency distribution of variables.

Null Hypothesis (H_0):

The null hypothesis states that the two categorical variables are independent of each other. In other words, there is no association between the variables.

H0:The variables are independent (no association)

Alternative Hypothesis (H_1):
The alternative hypothesis states that the two categorical variables are not independent of each other.
In other words, there is an association between the variables.

H1:The variables are not independent (there is an association)

Interpretation:

Reject H_0: If the calculated chi-square statistic is greater than the critical value
or if the p-value is less than or equal to the significance level (α),
you reject the null hypothesis.
This indicates that there is a significant association between the two categorical variables.

- **How many hours do you spend on social media?**

Males

H0: There is no significant difference between usage of social media and academic stream.
H1: There is a significant difference between usage of social media and academic stream.

	less than 1 hour	1 to 2 hour	3 to 4 hour	more than 4 hour	Total
Science	2	8	12	44	66
Commerce	2	6	6	1	15
Arts	2	4	4	11	21
Social science	3	11	16	11	41
Total	9	29	38	67	143

```
# Create the contingency table
social_media_data <- matrix(c(
  2, 8, 12, 44,
  2, 6, 6, 1,
```

```
  2, 4, 4, 11,
  3, 11, 16, 11
), nrow = 4, byrow = TRUE)


# Add row and column names
rownames(social_media_data) <- c("Science","Commerce","Arts","Social science")
colnames(social_media_data) <- c("<1 hour","1–2 hours","3–4 hours",">4 hours")

# Perform Chi-square test
chisq.test(social_media_data, simulate.p.value = TRUE)

Output:
X-squared = 28.592,  p-value = 0.0007591
```

P value=0.000759(<0.05)

Conclusion: There is a significant difference between usage of social media and academic stream.

- **Females.**

H0: There is no significant difference between usage of social media and academic stream.
H1: There is a significant difference between usage of social media and academic stream.

	less than 1 hr	1 to 2 hr	3 to 4 hr	more than 4hr	total
science	4	22	36	12	74
commerce	1	10	11	3	25
arts	4	11	12	3	30
social science	8		13	4	79
total	17	97	72	22	208

P value=0.002801(<0.05)

Conclusion: There is a significant difference between usage of social media and academic stream

- **What is your main purpose of using social media?**

Males

H0: There is no significant difference between purpose of using social media and academic stream.
H1: There is a significant difference between purpose of using social media and academic stream.

	Study	Time pass	Entertainment	To connect with people	Total
Science	12	25	10	9	56

Commerce	**7**	**4**	**2**	**2**	**15**
Arts	**6**	**7**	**4**	**4**	**21**
Social science	**15**	**17**	**3**	**6**	**41**
Total	**40**	**53**	**19**	**21**	**133**

```
# Create the contingency table (Males)

purpose_data <- matrix(c(
  12, 25, 10, 9,     # Science
   7,  4,  2, 2,     # Commerce
   6,  7,  4, 4,     # Arts
  15, 17,  3, 6      # Social science
),
nrow = 4,
byrow = TRUE)

# Add row and column names

rownames(purpose_data) <- c("Science", "Commerce", "Arts", "Social science")

colnames(purpose_data) <- c("Study", "Time pass", "Entertainment", "Connect with people")

# Perform Chi-square test

chisq_result <- chisq.test(purpose_data)

# Print result
print(chisq_result)
```

output:
X-squared = 7.0588,   P-value=0.63099961(>0.05)

Conclusion: There is no significant difference between purpose of using social media and academic stream.

- **Females**

H0: There is no significant difference between purpose of using social media and faculties.
H1: There is a significant difference between purpose of using social media and academic stream.

	Study	Time pass	Entertainment	To connect with people	Total
Science	**15**	**46**	**16**	**7**	**84**
Commerce	**15**	**9**	**1**	**0**	**25**
Arts	**12**	**17**	**1**	**0**	**30**
Social science	**35**	**28**	**11**	**5**	**79**
Total	**77**	**100**	**29**	**12**	**218**

P-value=0.00078991(<0.05)

Conclusion: There is a significant difference between purpose of using social media and academic stream.

T TEST FOR USAGE OF SOCIAL MEDIA IN DIFFERENT FACULTIES

- **How many hours do you spend on social media?**

For science faculty

H0: There is no significant difference between usage of social media and sex
H1: There is a significant difference between usage of social media and sex

```
# Perform independent t-test
# Scores for hour categories
scores <- c(0.5, 1.5, 3.5, 5) #mid value

# Male Science frequency counts
male_science <- c(2, 8, 12, 44)

# Female Science frequency counts
female_science <- c(4, 22, 36, 12)

# Expand data into individual-level values
male_hours <- rep(scores, male_science)
female_hours <- rep(scores, female_science)

# Perform independent t-test
t.test(male_hours, female_hours)
```

t = 5.1638, df = 135.84, p-value = 8.424e-07(<0.05)
95 percent confidence interval:
0.7282057 1.6321547
sample estimates:
mean of x mean of y
4.166667 2.986486

Conclusion: There is a significant difference between usage of social media and sex

For commerce

H0: There is no significant difference between usage of social media and sex
H1: There is a significant difference between usage of social media and sex

```
# Commerce Faculty

# Male Commerce frequency counts
```

```
male_commerce <- c(2, 6, 6, 1)

# Female Commerce frequency counts
female_commerce <- c(1, 10, 11, 3)

# Expand into individual-level values
male_hours <- rep(scores, male_commerce)
female_hours <- rep(scores, female_commerce)

# Perform independent t-test
t.test(male_hours, female_hours)
```

```
t = -0.81653, df = 28.886, p-value = 0.4209(>0.05)
95 percent confidence interval:
 -1.26188  0.54188
sample estimates:
mean of x mean of y
     2.40      2.76
```

Conclusion: There is no significant difference between usage of social media and sex

<u>For arts</u>

H0: There is no significant difference between usage of social media and sex
H1: There is a significant difference between usage of social media and sex

```
# Arts Faculty

# Male Arts frequency counts
male_arts <- c(2, 4, 4, 11)

# Female Arts frequency counts
female_arts <- c(1, 29, 12, 4)

# Expand into individual-level values
male_hours <- rep(scores, male_arts)
female_hours <- rep(scores, female_arts)

# Perform independent t-test
t.test(male_hours, female_hours)
```

```
output:
t = 3.1688, df = 29.853, p-value = 0.003522(<0.05)
95 percent confidence interval:
 0.467195 2.162205
sample estimates:
mean of x mean of y
 3.619048  2.304348
```

Conclusion: There is a significant difference between usage of social media and sex

For social science

H0: There is no significant difference between usage of social media and sex
H1: There is a significant difference between usage of social media and sex

```
# Social Science Faculty

# Male Social Science frequency counts
male_social <- c(3, 11, 16, 11)

# Female Social Science frequency counts
female_social <- c(11, 36, 13, 3)

# Expand into individual-level values
male_hours <- rep(scores, male_social)
female_hours <- rep(scores, female_social)

# Perform independent t-test
t.test(male_hours, female_hours)'
```

output:
t = 4.4412, df = 72.036, p-value = 3.166e-05(<0.05)
95 percent confidence interval:
0.6842963 1.7988628
sample estimates:
mean of x mean of y
3.146341 1.904762

Conclusion: There is a significant difference between usage of social media and sex

PIE CHART:
**Purpose of using Social Media by different faculties

Science

Purpose	Percentage
Study	19%
Entertainment	19%
Timepass	51%
To connect with people	11%

```
# Create data
purpose <- c("Study", "Entertainment", "Timepass", "To connect with people")
percentage <- c(47, 35, 10, 8)

# Create dataframe
df <- data.frame(purpose, percentage)

# Plot pie chart
pie(df$percentage,
   labels = df$purpose,
   main = "Purpose of Using Social Media")
```

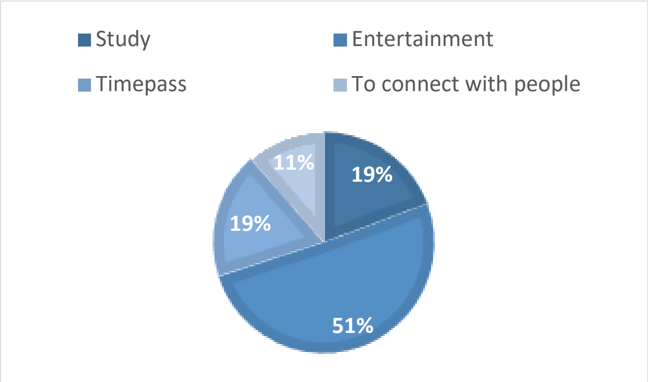

Figure1

Arts:

Purpose	Percentage
Study	47%
Entertainment	35%
Timepass	10%
To connect with people	8%

```
# Create data
purpose <- c("Study", "Entertainment", "Timepass", "To connect with people")
percentage <- c(47, 35, 10, 8)

# Create dataframe
df <- data.frame(purpose, percentage)

# Plot pie chart
```

```
pie(df$percentage,
    labels = df$purpose,
    main = "Purpose of Using Social Media")
```

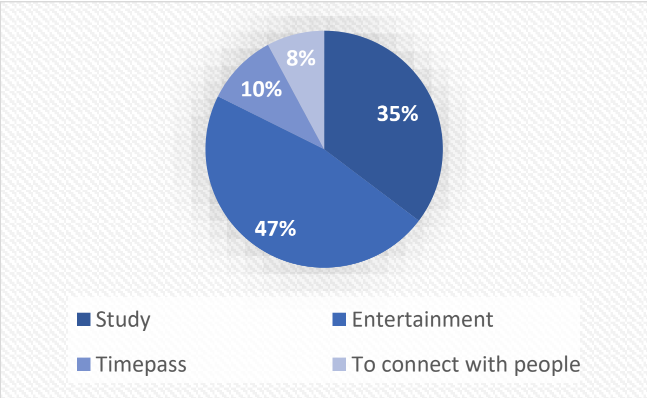

Figure2

Interpretation:
From the figure 2, we can see that 35.29% uses social media for study,47.05% for Entertainment,9.8% for Time pass and 7.84% for to connect with people

Commerce:

Purpose	Percentage
Entertainment	55%
Study	32%
Timepass	8%
To connect with people	5%

```
# Create data
purpose <- c("Study", "Entertainment", "Timepass", "To connect with people")
percentage <- c(55, 32, 8, 5)

# Create dataframe
df <- data.frame(purpose, percentage)

# Plot pie chart
pie(df$percentage,
    labels = paste(df$purpose, df$percentage, "%"),
    main = "Purpose of Using Social Media",
    col = rainbow(4))
```

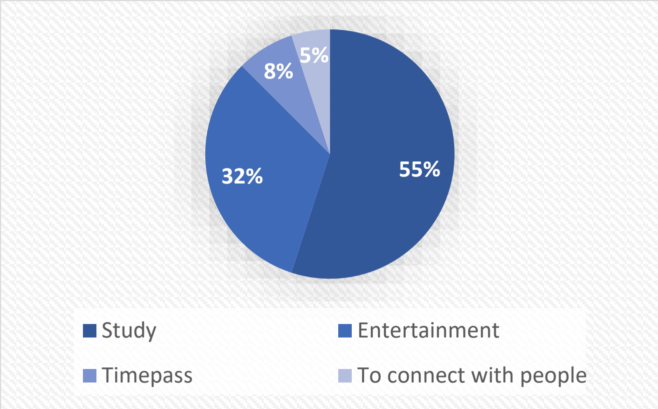

Figure3

Interpretation:
From the figure 3, we can see that 55% uses social media for study,32.5% for Entertainment,7.5% for Time pass and 5% for to connect with people

Social Science:

Platform	Percentage
Instagram	44%
Youtube	30%
Whatsapp	25%
Facebook	1%

```
# Create data
platform <- c("Instagram", "YouTube", "Whatsapp", "Facebook")
percentage <- c(44, 30, 25, 1)

# Create dataframe
df <- data.frame(platform, percentage)

# Plot pie chart
pie(df$percentage,
    labels = paste(df$platform, df$percentage, "%"),
    main = "Social Media Platforms Used by Social Science Students",
    col = c("pink", "skyblue", "lightgreen", "orange"))
```

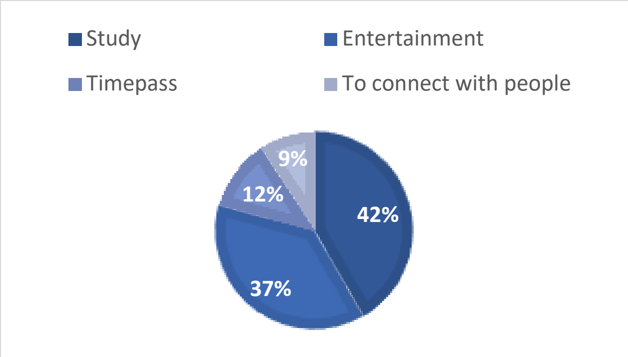

Figure4

Interpretation: From the figure 4, we can see that 41.66% uses social media for study,37.5% for Entertainment,11.66% for Time pass and 9.16% for to connect with people

MULTIPLE BAR DIAGRAM:

Social media Usage:

Faculty	< 1 hour	1–2 hours	3–4 hours	> 4 hours
Science	6	30	48	56
Commerce	3	16	17	5
Arts	6	15	17	14
Social Science	11	65	29	14

```
faculty <- c("Science", "Commerce", "Arts", "Social Science")

less_1 <- c(6, 3, 6, 11)
one_two <- c(30, 16, 15, 65)
three_four <- c(48, 17, 17, 29)
more_4 <- c(56, 5, 14, 14)

# Combine into matrix
data <- rbind(less_1, one_two, three_four, more_4)

# Plot multiple bar diagram
barplot(data,
        beside = TRUE,
        names.arg = faculty,
        col = c("orange", "green", "yellow", "brown"),
        xlab = "Faculty",
        ylab = "Number of Students",
        main = "Social Media Usage by Faculty")

# Add legend
legend("topright",
       legend = c("< 1 hour", "1–2 hours", "3–4 hours", "> 4 hours"),
       fill = c("orange", "green", "yellow", "brown"))
```

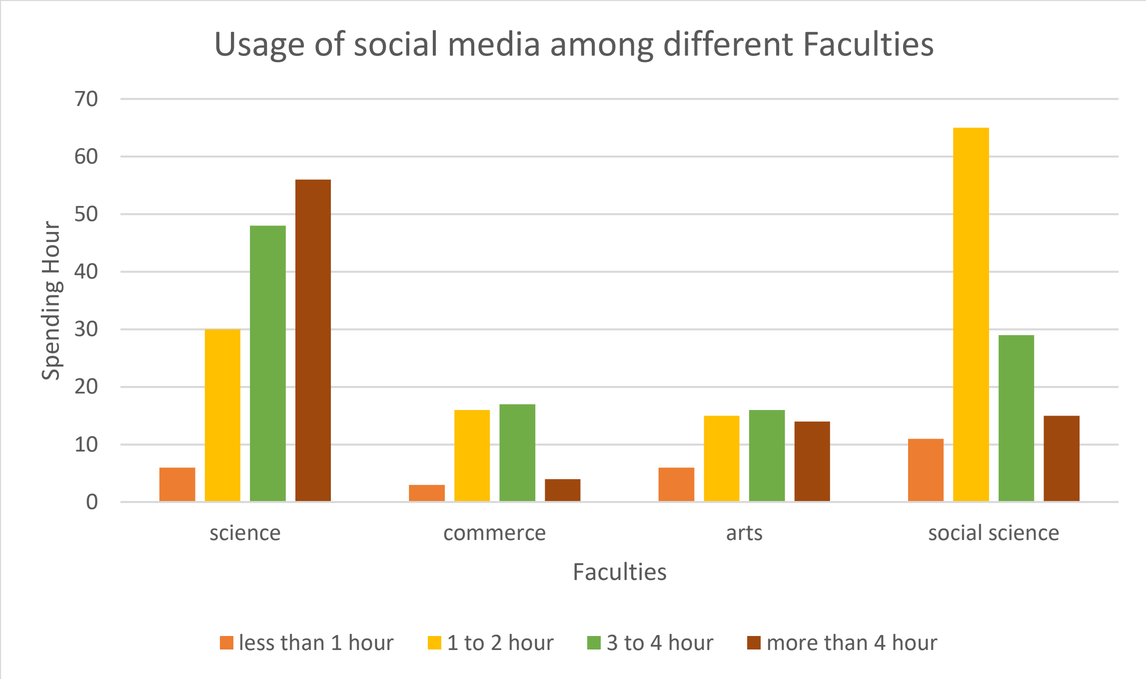

Figure 1

Interpretation: From the figure 1 it's clear that
*The maximum number of students spend more than 4 hours in Science Faculty
* The maximum number of students spend 3 to 4 Hours in commerce Faculty
* The maximum number of students spend 3 to 4 Hours in Arts Faculty
** The maximum number of students spend 1 to 2 Hours in Social Science Faculty

Negative impact of Social Media:

Faculty	Strongly Agree	Agree	Disagree	Strongly Disagree
Science	20	80	30	10
Arts	10	30	20	5
Commerce	5	40	10	5
Social Science	20	60	30	10

```
# Data
faculties <- c("Science", "Arts", "Commerce", "Social Science")

strongly_agree <- c(20, 10, 5, 20)
agree <- c(80, 30, 40, 60)
disagree <- c(30, 20, 10, 30)
strongly_disagree <- c(10, 5, 5, 10)

# Combine into matrix
data <- rbind(strongly_agree, agree, disagree, strongly_disagree)

# Plot grouped bar chart
```

```
barplot(data,
        beside = TRUE,
        names.arg = faculties,
        col = c("darkgreen","skyblue","orange","red"),
        xlab = "Faculty",
        ylab = "Count",
        main = "Negative Impact of Social Media")

# Add legend
legend("topright",
       legend = c("Strongly Agree", "Agree", "Disagree", "Strongly Disagree"),
       fill = c("darkgreen","skyblue","orange","red"))
```

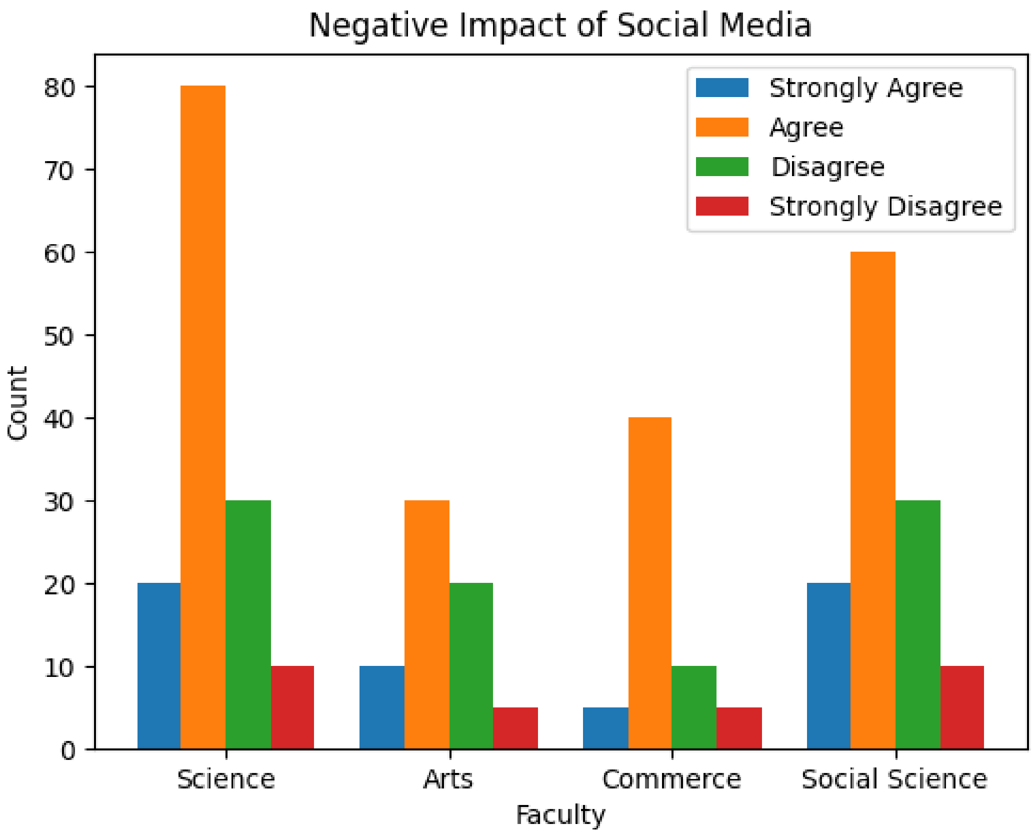

Figure 2

Interpretation : From the figure 2 it's clear that
All faculty students agree that Social media is negatively impacts mental health more than positive impact

QUESTIONNAIRE ON EFFECT OF SOCIAL MEDIA ON SUDENTS MENTAL HEALTH

- Name of the student:

  ____

- Place of residence:
  - Rural
  - Urban

- Sex:
  - Male
  - Female

- Family Income:
  - Below 2 lakh
  - between 2 and 3 lakh
  - between 3 and 4 lakh
  - above 4 lakhs

- Qualification of the father
  - Uneducated
  - SSLC/PUC
  - Graduation
  - Post graduation

- Qualification of the mother
  - Uneducated
  - SSLC/PUC
  - Graduation
  - Post graduation

- Occupation of the father:
  - Government job
  - Private job
  - Agriculture
  - Labour

- Occupation of the mother:
  - Government job
  - Private job
  - Housewife

- others

1.How many hours do you spend on social media?
- Less than one hour per day
- 1 to 2 hours per day
- to 4 hours per day
- More than 4 hours per day

2.Which platform do you use regularly?
- Facebook
- Instagram
- Youtube
- Whatsapp

3.What is your main purpose of using social media?
- Study
- Entertainment
- Time pass
- to connect with people

4.Do you feel anxious or stressed when you don’t have access to social media?
- Frequently
- More frequently
- Less frequently
- No

5.Has social media ever negatively affected your self esteem?
- Yes
- No

6.Do you feel addicted to social media?
- Yes
- No

7.What activities do you often neglect because of social media?
- Studying
- Sleeping
- Socializing with friends
- Other activities

8.Have you experienced cyberbullying on social media?
- Yes

- No

9.Does social media enhance your overall well-being and happiness?

- Yes
- No

10.Have you ever taken a break from social media to improve your mental well-being?

- Frequently
- More frequently
- Less frequently
- No

11.Do you think social media has improved your relationships?

- Yes
- No

12.Which emotions do you typically experience while scrolling through social media?

- Happiness
- Sadness
- Inspiration
- Inadequacy

13.Do you agree that social media negativity impacts mental health more than positively?

- strongly agree
- agree
- disagre
- strongly disagree
  s

14.Do you experience sleep disturbances due to social media

- Yes
- No

15.Do you believe you are addicted to online gaming?

- Yes
- No

16.Have you spent more money than intended on online games?

- strongly agree
- agree
- disagree
- strongly disagree

17.Has social media affected your academic performance?

- Yes
- No

18.How do you feel about the content you see on social media?

- useful
- not useful